

Integral Relations among Totient Sections

Finite-level bases and bounded residue series

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ABSTRACT

For integers $k \geq 2$ and $e \geq 1$, every totient section $n \mapsto \varphi(k^j n + r)$ with $0 \leq j \leq e, 0 \leq r < k^j$, is an explicit integer multiple of one of $k^e + 1$ independent sections. The resulting two-term relations form an integral basis of all relations among the indexed sections: their coefficients are recovered without division. Martin's stronger affine theorem gives the independence; we also prove it directly by making an evaluation matrix diagonal modulo a prime. The same separation method classifies rational binary series formed from dyadic totient residues. It does not settle the irrationality of the unreduced binary totient series.

1 Introduction

Splitting the totient sequence into its even and odd subsequences, then splitting once more, gives seven sequences:

$$\varphi(n); \quad \varphi(2n), \varphi(2n+1); \quad \varphi(4n), \varphi(4n+1), \varphi(4n+2), \varphi(4n+3).$$

The identities

$$\varphi(4n) = 2\varphi(2n), \quad \varphi(4n+2) = \varphi(2n+1)$$

leave five independent sequences. Thus every identity among the seven comes from these two substitutions. More precisely, an integral relation is a unique integral combination of the two displayed relations. We will prove the same statement at every depth and in every integer base.

For an integer $k \geq 2$, the *totient sections* are the sequences $n \mapsto \varphi(k^j n + r)$, where $j \geq 0$ and $0 \leq r < k^j$. Throughout, n ranges over $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\varphi(0) = 0$. To describe relations we retain a separate index for each pair (j, r) , even when two pairs give the same sequence. For example, $\varphi(4n+2)$ and $\varphi(2n+1)$ are equal sequences but occupy different positions in the list above.

The sections form the k -kernel of the totient sequence. A sequence is k -regular when the $\mathbb{Z}$ -module generated by its k -kernel is finitely generated [1]. Coons proved that the totient sequence is not k -regular for any $k \geq 2$ [2, Thm. 3.2]. Here we determine those modules at each finite level, including their integral relations. The theorem gives the basis and the substitutions; Corollary 3.1 explains how to recover every relation from them. Level zero contains only $\varphi(n)$ and has dimension one.

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Theorem 1.1 (A basis through each finite level). *Let $k \geq 2$ and $e \geq 1$ be integers, write $F_{j,r}^{(k)}(n) = \varphi(k^j n + r)$, and put*

$$V_{k,e} = \text{span}_{\mathbb{Q}} \{ F_{j,r}^{(k)} : 0 \leq j \leq e, 0 \leq r < k^j \}.$$

Then $\dim_{\mathbb{Q}} V_{k,e} = k^e + 1$, and

$$\mathcal{B}_{k,e} = \{F_{0,0}^{(k)}, F_{1,0}^{(k)}\} \cup \{F_{j,r}^{(k)} : 1 \leq j \leq e, 1 \leq r < k^j, k \nmid r\}$$

is a basis. Every omitted section reduces to a basis element by an explicit scalar: $F_{j,0}^{(k)} = k^{j-1} F_{1,0}^{(k)}$ for $j \geq 1$, and if $r = k^t u$ with $t = \max\{s : k^s \mid r\} \geq 1$ and $k \nmid u$, then

$$F_{j,r}^{(k)} = C_k(t, u) F_{j-t, u}^{(k)}, \quad C_k(t, u) = k^t \prod_{\substack{p \mid k \\ p \nmid u}} \left(1 - \frac{1}{p}\right).$$

The condition $k \nmid r$ allows residues with $\gcd(k, r) > 1$. For instance, $r = 1$ and $r = 2$ are both retained in base six, and the reductions involving them are

$$\varphi(36n + 6) = 2\varphi(6n + 1), \quad \varphi(36n + 12) = 4\varphi(6n + 2).$$

The factor 2 is already present in $6n + 2$ but not in $6n + 1$; only newly introduced prime divisors contribute factors $1 - 1/p$. Thus one removes whole factors of k from the residue, not every common prime factor. Repeated prime factors of k cause no exception.

Martin’s 2006 theorem on simultaneous totient inequalities already implies independence for every finite family of pairwise nonproportional affine forms with positive integral slopes [4, Thm. 1]. Its conclusion is stronger than the independence used here. The point of the present description is the explicit reduction of the sections and the integral relation basis in Corollary 3.1. We also give a direct proof that constructs a nonsingular evaluation matrix. It uses the CRT–Dirichlet separation argument in Yazdani’s proof of Theorem 2, credited there to Shallit [12, pp. 652–653]. The step that needs care is to keep the progression for the selected affine form reduced while imposing prime divisors on all the others.

The proof in Section 2 separates scalar reduction from independence. Section 3 recovers the integral relations, starting in base six. Section 4 uses prime isolation for bounded functions of totient residues. The series

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$$

has unbounded coefficients. Its irrationality, asked in Erdős Problem #249 [10, p. 61], remains open. The coefficient basis supplies no irrationality conclusion for this value. Appendix A gives further coefficient consequences; Appendices B and C collect the tail identities, comparison sequences and still-unproved cancellation estimates relevant to S . Their longer proofs are in *The Binary Totient Series*.

2 Proof of the basis theorem

Scalar reductions

The product formula proves spanning. For independence we first separate nonproportional affine arguments, then handle the two proportional arguments n and kn separately. The reductions depend only on which prime divisors of k are already present. For a zero residue, kn and $k^j n$ have the same prime divisors when $n > 0$. Hence $\varphi(k^j n) = k^{j-1} \varphi(kn)$; both sides vanish at $n = 0$. For an omitted positive residue $r = k^t u$ with $t \geq 1$ and $k \nmid u$, we have $t < j$ because $0 < r < k^j$. Put $m = k^{j-t} n + u$. A prime $p \mid k$ divides m exactly when it divides u . The product formula for $\varphi(k^t m)/\varphi(m)$ therefore contributes k^t and one factor $1 - 1/p$ for each prime $p \mid k$ absent from u . The multiplier is an integer: its denominator is a product of distinct primes dividing k , and therefore divides k^t . This gives $C_k(t, u)$ and proves that the stated family spans $V_{k,e}$.

Affine independence

Independence follows from Martin's theorem. For any ordering of pairwise nonproportional affine forms and any $C > 0$, it gives a set of positive lower density on which each successive totient ratio exceeds C . In a proposed nonzero relation, place a nonzero coefficient c_1 first and take $C > \max(1, \sum_{i>1} |c_i|/|c_1|)$. The first term then dominates the sum of all remaining terms in absolute value, a contradiction [4, Thm. 1]. This also proves independence after any finite prefix is deleted.

The direct proof separates one column at a time: all its competitors will vanish modulo a prime, while its own entry will not. Let $L_i(n) = a_i n + b_i$ for $1 \leq i \leq s$, where $a_i > 0$ and $a_i b_j - a_j b_i \neq 0$ for $i \neq j$. A shift of n makes all the b_i positive and leaves these cross determinants unchanged. Write $g_i = \gcd(a_i, b_i)$ and $A_i = L_i/g_i$. We must make A_i prime, not L_i : the fixed divisor g_i may prevent L_i from taking large prime values. We seek a prime ℓ and points $n_1, \dots, n_s$ for which

$$E = (\varphi(L_j(n_i)))_{1 \leq i, j \leq s} \equiv \begin{pmatrix} \varphi(g_1) & 0 & \cdots & 0 \\ 0 & \varphi(g_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \varphi(g_s) \end{pmatrix} \pmod{\ell}, \quad \ell \nmid \prod_i \varphi(g_i). \quad (1)$$

Evaluating a rational relation at these points gives $Ec = 0$. But $\det E \not\equiv 0 \pmod{\ell}$, hence E is invertible over $\mathbb{Q}$ and $c = 0$. It remains to construct its rows.

For row i , we require $L_j(n_i)$ to have a prime divisor $q_{ij} \equiv 1 \pmod{\ell}$ whenever $j \neq i$. Then $q_{ij} - 1$ divides $\varphi(L_j(n_i))$, giving the off-diagonal zeros in (1). For the diagonal entry, we require $A_i(n_i)$ to be a prime congruent to 2 modulo ℓ . Choose an odd ℓ dividing none of the slopes a_i , the products $g_i \varphi(g_i)$ or the nonzero cross determinants. For each fixed i , Dirichlet's theorem supplies distinct primes $q_{ij} \equiv 1 \pmod{\ell}$ ($j \neq i$) outside the same finite exceptional set. We impose

$$A_i(n) \equiv 2 \pmod{\ell}, \quad L_j(n) \equiv 0 \pmod{q_{ij}} \quad (j \neq i).$$

Each congruence is soluble because its slope is coprime to its modulus. CRT combines them into $n = n_0 + Ru$, where $R = \ell \prod_{j \neq i} q_{ij}$.

CRT supplies the congruences, but not yet a prime value. Dirichlet applies to the progression of values of A_i , namely

$$A_i(n_0 + Ru) = A_i(n_0) + (a_i/g_i)Ru.$$

Its initial value must be coprime to its step. The step has only three kinds of prime factors: those of a_i/g_i , excluded by primitivity; ℓ , excluded by the residue 2; and the imposed q_{ij} . Such a q_{ij} cannot divide $A_i(n_0)$: it would then divide both $L_i(n_0)$ and $L_j(n_0)$, hence

$$a_i L_j(n_0) - a_j L_i(n_0) = a_i b_j - a_j b_i,$$

contrary to its choice. This is where nonproportionality is used: the nonzero cross determinant lets us keep each imposed divisor away from the selected form. We may now choose $p = A_i(n_i)$ prime, as large as needed, and in particular with $p \nmid g_i$. Then

$$\varphi(L_i(n_i)) = \varphi(g_i)(p-1) \equiv \varphi(g_i) \not\equiv 0 \pmod{\ell},$$

while all the other entries of row i vanish modulo ℓ . We choose the rows separately, applying Dirichlet's theorem once more for each selected form A_i . Thus no simultaneous primality assertion is required. The construction proves (1) with every n_i beyond any prescribed threshold.

The two zero-residue sections

The arguments n and kn are proportional, so they escape the preceding argument. On $n = km + 1$ we have $\varphi(kn) = \varphi(k)\varphi(n)$: this merges their coefficients and lets affine independence remove all the other sections. The positive-residue sections become the forms

$$k^{j+1}m + k^j + r \quad (k \nmid r),$$

together with $km + 1$. These are pairwise nonproportional: equality of two intercept-to-slope ratios would give $r/k^j = r'/k^{j'}$, which for $j' > j$ forces $k \mid r'$. No positive-residue form is proportional to $km + 1$. If A, B multiply $\varphi(n), \varphi(kn)$, affine independence therefore leaves $A + B\varphi(k) = 0$ and eliminates every other coefficient. At $n = k$, however, the ratio is k , giving $A + Bk = 0$. Since $\varphi(k) < k$, these two equations separate the remaining coefficients: $A = B = 0$. Finally,

$$|\mathcal{B}_{k,e}| = 2 + \sum_{j=1}^e (k^j - k^{j-1}) = k^e + 1,$$

which completes the proof of Theorem 1.1. For independence on a tail, use a sufficiently large power of k in the last evaluation.

3 Integral relations

Once independence is known, a linear identity can be tested by substitution alone: reduce to the retained sections and compare their coefficients. Consider base six at level two, where the family has 43 indices. The 37 retained sections are $F_{0,0}, F_{1,0}$, the five $F_{1,r}$

($1 \leq r \leq 5$), and the 30 sections $F_{2,r}$ with $1 \leq r < 36$ and $6 \nmid r$. The omitted sections satisfy

$$\begin{aligned} F_{2,0} &= 6F_{1,0}, & F_{2,6} &= 2F_{1,1}, & F_{2,12} &= 4F_{1,2}, \\ F_{2,18} &= 3F_{1,3}, & F_{2,24} &= 4F_{1,4}, & F_{2,30} &= 2F_{1,5}. \end{aligned}$$

The substitutions test an identity by leaving 37 retained coefficients to compare with zero. To recover the identity itself, we must distinguish the formal symbols $E_{j,r}$ from the sequences $F_{j,r}$ they represent: the symbols are independent even when the sequences coincide. For example, the substitution $F_{2,12} = 4F_{1,2}$ corresponds to the decomposition

$$x_{1,2}E_{1,2} + x_{2,12}E_{2,12} = (x_{1,2} + 4x_{2,12})E_{1,2} + x_{2,12}(E_{2,12} - 4E_{1,2}).$$

The retained coefficient becomes $x_{1,2} + 4x_{2,12}$, while the coefficient of the removed relation is the original $x_{2,12}$. Repeating at the six omitted indices both tests the identity and recovers its unique relation coefficients. In the opening binary example the relation vectors are $E_{2,0} - 2E_{1,0}$ and $E_{2,2} - E_{1,1}$. The same calculation gives the general result.

Corollary 3.1 (Integral coordinates and all integral relations). *Let $k \geq 2$ and $e \geq 1$. The retained family is a $\mathbb{Z}$ -basis of the module generated by the sections through level e . Index the sections by their level and residue, retaining distinct indices even when they define equal sequences. For each omitted index i , write the scalar reduction as $F_i = a_i F_{j(i)}$, where $j(i)$ is retained and a_i is a nonnegative integer. In the free abelian group with one generator E_i for each of these indices, the vectors*

$$R_i = E_i - a_i E_{j(i)} \quad (i \text{ omitted})$$

form a $\mathbb{Z}$ -basis of the kernel of evaluation $E_i \mapsto F_i$. In particular, every integral relation has a unique integral expression in these elementary relations, and their rank is $\sum_{j=1}^{e-1} k^j$.

Proof. Since $\prod_{p|k, p \nmid u} p$ divides k , every scalar $C_k(t, u)$ is an integer. The retained family therefore spans the section module over $\mathbb{Z}$, and its rational independence makes it a $\mathbb{Z}$ -basis.

For any vector $x = \sum_i x_i E_i$, substitute $E_i = R_i + a_i E_{j(i)}$ at each omitted index and collect terms:

$$x = \sum_{j \text{ retained}} \left(x_j + \sum_{\substack{i \text{ omitted} \\ j(i)=j}} a_i x_i \right) E_j + \sum_{i \text{ omitted}} x_i R_i.$$

The first sum gives the retained coordinates of the evaluated sequence; the second evaluates to zero. Thus x is a relation exactly when the first sum vanishes, in which case $x = \sum_{i \text{ omitted}} x_i R_i$. This expression is unique: at omitted index i , the coefficient of $\sum_h c_h R_h$ is c_i itself. Each substitution has a unit coefficient at its own omitted index, so the entire calculation takes place over $\mathbb{Z}$, without division. Counting omitted indices gives $(1 + k + \dots + k^e) - (k^e + 1) = \sum_{j=1}^{e-1} k^j$. $\square$

This is an integral basis of the *section-generated* module, not of every integer-valued sequence in its rational span. For $k = 2$, $e \geq 2$, the sequence $\varphi(4n + 3)/2$ is integer-valued but has retained coordinate $1/2$. Thus the section module is not saturated in

the ambient group of integer-valued sequences. There is no conflict with the integral relation basis: that basis concerns relations among the given generators, not divisibility of the sequences they generate.

An alternative is to test equality at finitely many inputs. Return to base six and level two. Order the retained columns as above, with residues increasing within each level, and evaluate at $n = 0, \dots, 36$ to obtain a 37×37 matrix. Its determinant is 17 modulo 101, so it is nonsingular over $\mathbb{Q}$. Consequently a retained combination is zero if and only if all 37 of its integer evaluations are zero. Here reduction modulo 101 certifies the matrix, not the equality of a proposed combination: $101F_{2,2}$ would vanish modulo 101 at every input, although its integer value at $n = 0$ is 101. The script `scripts/totient_kernel_normal_form.py` calculates the coordinates and the first nonzero evaluation using integer arithmetic.

4 Bounded functions of totient residues

The matrix argument separates one value from finitely many others. Now we separate one coefficient from all its neighbours in a long window. Boundedness turns that local separation into an irrationality proof: the right zero block forces a rational tail to vanish exactly; the left makes the isolated nonzero term too small to be rational with the same denominator. Residues in the first sum below are least nonnegative representatives, not necessarily binary digits.

Theorem 4.1 (Series of totient residues). *For every $m \geq 3$,*

$$\sum_{n \geq 1} \frac{\varphi(n) \bmod m}{2^n} \notin \mathbb{Q}.$$

For $k \geq 1$ and $f : \mathbb{Z}/2^k\mathbb{Z} \rightarrow \mathbb{Q}$, the series $\sum_{n \geq 1} f(\varphi(n) \bmod 2^k) 2^{-n}$ is rational exactly when f is constant on the even residue classes. If that constant is c , its value is $3f(1)/4 + c/4$.

Only $f(1)$ and the even classes are visible to the series: $\varphi(1) = \varphi(2) = 1$ and all later totients are even. This explains the rational case, including the value $3/4$ modulo two. The irrational case rests on the pattern

$$\underbrace{0, \dots, 0}_{L \text{ terms}}, \quad a_N \neq 0, \quad \underbrace{0, \dots, 0}_{L \text{ terms}}.$$

The coefficients outside the window may have either sign; only their uniform bound matters.

Lemma 4.2 (An isolated nonzero coefficient between long zero blocks). *Let $a_n \in \mathbb{Z}$ satisfy $|a_n| \leq C$. Suppose that for arbitrarily large L there is $N > L$ such that $a_N \neq 0$ and $a_{N+t} = 0$ for $0 < |t| \leq L$. Then $\sum_{n \geq 1} a_n 2^{-n}$ is irrational.*

Proof. Suppose that the sum has denominator $q \geq 1$. Multiplying it by 2^j and removing its finite integer prefix shows that every scaled tail $T_j = \sum_{t \geq 1} a_{j+t} 2^{-t}$ belongs to $q^{-1}\mathbb{Z}$. Choose a pattern with $C2^{-L} < 1/q$. The block on the right gives

$$|T_N| \leq C \sum_{t > L} 2^{-t} = C2^{-L} < 1/q.$$

The only element of $q^{-1}\mathbb{Z}$ in this interval is zero, so $T_N = 0$. The right block has removed the unknown tail exactly. Starting before the left block now leaves only the nonzero centre, rescaled:

$$T_{N-L-1} = a_N 2^{-L-1}, \quad 0 < |T_{N-L-1}| \leq C 2^{-L-1} < 1/q,$$

contradicting $T_{N-L-1} \in q^{-1}\mathbb{Z}$. □

For example, the indicator of the squares satisfies the lemma. An eventually periodic sequence with infinitely many nonzero terms does not have isolated coefficients at arbitrarily large distances. The uniform bound C is essential to this argument.

Proof of Theorem 4.1. We prescribe a prime at the centre and a different divisor at each neighbour. Fix a unit $s \pmod{m}$ and an integer $L \geq 1$, and choose distinct primes $q_t \equiv 1 \pmod{m}$, all exceeding L , for $0 < |t| \leq L$. The congruences

$$p \equiv s \pmod{m}, \quad p \equiv -t \pmod{q_t} \quad (0 < |t| \leq L)$$

define a reduced progression, since s is a unit modulo m and $0 < |t| < q_t$. CRT and Dirichlet's theorem therefore allow us to choose an arbitrarily large prime $p > L + 1$ on this progression, with $m \mid \varphi(p + t)$ at every neighbour and $\varphi(p) \equiv s - 1 \pmod{m}$ at the centre.

For a rational-valued function f on the residue classes, subtract $f(0)$ and multiply by a common denominator of its values. Since $\sum_{n \geq 1} 2^{-n} = 1$, these operations preserve whether the sum is rational. If $f(s - 1) \neq f(0)$, the resulting bounded integer sequence has an isolated nonzero coefficient for every L , so the lemma applies. For the least-residue map, $s = -1$ gives the nonzero centre $m - 2$ when $m \geq 3$. For $m = 2^k$, the possible centres $s - 1$ run through *every* even class. Thus any failure of constancy on those classes supplies a centre with $f(s - 1) \neq f(0)$. Conversely, constancy makes every term after $n = 2$ equal c , giving $3f(1)/4 + c/4$. □

The dyadic modulus matters to the classification. Modulo 6, for example, prime centres of this form see only $s - 1 \in \{0, 4\}$, not the even class 2. The least-residue assertion needs just one nonzero centre; classification of an arbitrary function needs all relevant classes.

The first assertion is also an application of Erdős's sparse-series criterion [6, Lem. 1, p. 213], and of Kaneko, Suzuki and Tachiya [7, Thm. 3]. The residues are bounded and nonnegative, and the prime construction above makes their support infinite. The nonzero residues occur on a set of density zero: an integer in that set has no prime divisor $p \equiv 1 \pmod{m}$. Wong gave the case in which the base equals the modulus [5]. The isolated-coefficient argument above also treats signed values of $f - f(0)$, for which those nonnegative sparse-series criteria need an additional spacing hypothesis. The [extension to other bases](#) treats every integer $t \geq 2$ and proves rational linear independence of 1 and $\sum_n (\varphi(n) \bmod m) t^{-dn}$, $d \geq 1$.

A Further consequences for the sections

Every finite set of dyadic sections lies in a single finite level. Theorem 1.1 therefore also determines the full kernel; “span” and “relation” here still mean finite linear combinations.

Corollary A.1 (Dyadic basis). *The family*

$$\{\varphi_{0,0}, \varphi_{1,0}\} \cup \{\varphi_{j,r} : j \geq 1, 0 < r < 2^j, r \text{ odd}\}, \quad \varphi_{j,r}(n) = \varphi(2^j n + r),$$

is a basis for the rational span of the full dyadic kernel. Every rational relation is generated by the scalar reductions. The complete level-zero truncation has dimension one.

Proof. Each finite subset of this family is contained in some $\mathcal{B}_{2,e}$. Theorem 1.1 gives independence, and its reductions give spanning. At level zero the sole section $\varphi(n)$ is nonzero, since $\varphi(1) = 1$. $\square$

The affine argument also permits periodic coefficients by restriction to their common residue classes.

Corollary A.2 (Periodic coefficients). *Let $L_1, \dots, L_s$ be pairwise nonproportional affine forms with integer coefficients and positive slopes. If $w_1, \dots, w_s$ are rational-valued periodic sequences, then*

$$\sum_i w_i(n) \varphi(L_i(n)) = 0 \quad \text{for all sufficiently large } n \quad \implies \quad w_i(n) = 0 \quad \text{for every } i, n.$$

Proof. We choose a common period Q and restrict to $n = r + Qt$. The coefficients are now the constants $w_i(r)$, and the cross determinants are $Q(a_i b_j - a_j b_i) \neq 0$. By shifting t beyond the given threshold we can apply affine independence, obtaining $w_i(r) = 0$ for every i . We repeat this for each residue r ; periodicity then gives the conclusion even at the indices preceding the threshold. $\square$

Proportional forms are excluded for a reason: the identity $\varphi(4n + 2) = \varphi(2n + 1)$ would otherwise contradict the conclusion. Martin’s theorem gives the same corollary on each progression [4, Thm. 1].

Coons proves nonregularity using the poles of $\zeta(s - 1)/\zeta(s)$ [2, Cor. 3.1, Thm. 3.2]. The ranks above recover it: finite rational kernel dimension would embed the integral kernel module by evaluations in $\mathbb{Z}^d$, making it finitely generated; the converse is immediate (compare [3, App. A]). Allouche, Shallit and Yassawi survey the earlier separation method [13, Thm. 3, Ex. 4, Rem. 7].

The generating function $F_\varphi(z) = \sum_{n \geq 1} \varphi(n) z^n$ therefore satisfies no k -Mahler equation $\sum_{i=0}^d P_i(z) F_\varphi(z^{k^i}) = 0$ with $P_i \in \mathbb{Q}[z]$, $P_0 \neq 0$. Indeed, $\varphi(n) \leq n$ gives logarithmic height at most $\log n$, and the height theorem of Adamczewski, Bell and Smertnig would make such a series k -regular [16, Thm. 1.2(b)]. Bell and Smertnig’s multiplicative classification gives the same conclusion [14, Thm. 1.3]. Bell, Bruin and Coons treat related algebraic and D -finite series (those satisfying polynomial-coefficient linear differential equations) [15, Thms. 1.5–1.6]. These are statements about the function, not its value at $1/2$.

The section operators $T_i g(n) = g(kn + i)$ satisfy

$$T_i F_{j,r} = F_{j+1,ki+r}, \quad V_{k,e+1} = V_{k,e} + \sum_{i=0}^{k-1} T_i V_{k,e}.$$

Every residue at level $j+1$ has the form $k^j i + r$, giving the second identity. A positive retained residue stays retained. For the zero-residue generators, reduce $T_i F_{0,0} = F_{1,i}$ and $T_i F_{1,0} = F_{2,ki}$ as before. Each image is a positive integer multiple of a basis element, but rank growth prevents a finite invariant truncation. Consequently the growth formula using the joint spectral radius for regular sequences [3, Thm. 1, Prop. 4, Cor. 7] does not apply.

B The binary totient series

The proof of Theorem 4.1 uses boundedness where a long zero block makes the tail small. For S , the estimate $0 \leq \varphi(n) \leq n$ proves convergence, but forcing $2^L \mid \varphi(N+L)$ with $N+L > 1$ already requires $2^L \leq N+L-1$. The tail bound $(N+L+2)2^{-L}$ then exceeds 1. This is a failure of the estimate, not a lower bound on the tail. The same divisibility that creates the gap can therefore move it too far out for this bound to be useful. We turn instead to a finite test for differences of scaled tails.

Put $R_N = \sum_{j \geq 1} \varphi(N+j)2^{-j}$ and $\Delta_h(N) = R_{N+h} - R_N$. Subtracting the finite-prefix identities for $2^{N+h}S$ and $2^N S$ gives

$$\Delta_h(N) \equiv 2^N(2^h - 1)S \pmod{\mathbb{Z}}. \quad (2)$$

If $S = a/(2^c v)$, with v odd, then $\Delta_h(N)$ is integral for $N \geq c$ and $h = \varphi(v)$. The tails are then periodic modulo integers, not necessarily equal. To disprove rationality it therefore suffices to find a nonintegral difference beyond every threshold for each positive h .

For $h, N, L \in \mathbb{N}$, define the integer

$$D_{h,N,L} = \sum_{j=0}^{L-1} (\varphi(N+h+1+j) - \varphi(N+1+j))2^{L-1-j}, \quad B = N+h+L+2.$$

Since $0 \leq R_M \leq M+2$, we have

$$2^L \Delta_h(N) - D_{h,N,L} = R_{N+h+L} - R_{N+L}, \quad |2^L \Delta_h(N) - D_{h,N,L}| \leq B. \quad (3)$$

Nonnegativity gives the larger tail bound, not the sum. Thus $D_{h,N,L}/2^L$ locates $\Delta_h(N)$ within $B/2^L$. The error interval contains no integer precisely when the residue stays more than B from both endpoints:

$$B < D_{h,N,L} \bmod 2^L < 2^L - B. \quad (4)$$

This proves $\Delta_h(N) \notin \mathbb{Z}$ and, by (2), excludes every reduced denominator dividing $2^N(2^h - 1)$. For example, $D_{1,12,16} = -143140$ has residue 53468 modulo 2^{16} , and $B = 31$, so it excludes denominators dividing 2^{12} . In ordinary coordinates the same certificate reads

$$\Delta_1(12) \in \left[\frac{-143171}{65536}, \frac{-143109}{65536} \right] \subset (-3, -2).$$

The test is complete for a *fixed* difference: if $\delta = \text{dist}(\Delta_h(N), \mathbb{Z}) > 0$, eventually $2^L \delta > 2B$, since $B = N + h + L + 2$ grows linearly. The same error estimate gives (4). Failure at one depth proves no integrality; the interval may be too wide. Increasing L detects an existing nonintegral difference, but does not supply one for each h beyond every threshold.

B.1 Equivalent residue conditions

We write $K(h, N, L)$ for (4).

Theorem B.1 (Propagation of one nonintegral tail difference). *Fix $d \geq 1$ and $N \geq 0$. If $\Delta_d(N) \notin \mathbb{Z}$, then every sufficiently late pair $\{t, t+1\}$ contains an m such that $K(md, N, md)$ holds. Consequently*

$$\exists t \geq 1 : K(td, N, td) \iff \Delta_d(N) \notin \mathbb{Z}.$$

Requiring this for every $d \geq 1, N \geq 0$ is equivalent to $S \notin \mathbb{Q}$.

Completeness alone lets the depth grow while the shift stays fixed. Here the two must grow together. The recurrence $\Delta_{(t+1)d}(N) \equiv 2^d \Delta_{td}(N) + \Delta_d(N) \pmod{\mathbb{Z}}$ provides the extra step: two consecutive failures would make the fixed nonintegral $\Delta_d(N)$ arbitrarily close to an integer. The estimates are in [Appendix A of the companion paper](#). A single pair d, N need not imply irrationality; the final assertion requires every positive d and every $N \geq 0$.

For a hypothetical denominator $2^c v$, a second test uses the modulus $M = (2^H - 1)/v$, where H is a positive multiple of $\varphi(v)$. Write $B_{H,c} = \sum_{j=0}^{H-1} \varphi(c+1+j)2^{H-1-j}$.

Equivalent formulation B.2 (A residue test modulo a divisor of $2^H - 1$). The series S is irrational if and only if, for every $c \geq 0$ and odd $v \geq 1$, some positive multiple H of $\varphi(v)$ satisfies

$$c + H + 1 < (-B_{H,c}) \bmod M < M - (c + H + 1). \quad (5)$$

The identity $B_{H,c} = M(vR_c) - \Delta_H(c)$ explains the modulus. Under rationality, $vR_c \in \mathbb{Z}$, so only the small tail difference remains modulo M . Conversely, for irrational S , the distance of the fixed number vR_c from $\mathbb{Z}$ is positive; multiplication by the exponentially growing M eventually dominates the linear error. The [proof and residue recurrence](#) make this comparison explicit. The required residues have not been shown to exist for every choice of the parameters in either test. A separate [Farey-interval calculation](#) at depth 240 excludes $S = a/q$ for $0 < q \leq Q_0 \approx 7.96 \times 10^{34}$.

B.2 Comparison sequences

To turn section rank into irrationality, one would need an upper bound on carry rank under rationality. The following comparison shows why rationality and linear growth alone cannot give it. There is an integer sequence c with $0 \leq c(n) \leq n$, unchanged odd totients and $|c(n) - \varphi(n)| \leq 2$ at even indices, for which $\sum c(n)2^{-n} = 5/4$. Its carry is the integer sequence

$$u_N = 4 \sum_{j \geq 1} c(N+j)2^{-j} = 5 \cdot 2^N - 4 \sum_{n=1}^N c(n)2^{N-n}.$$

Its recurrence is $2u_{N-1} - u_N = 4c(N)$ for $N \geq 1$. For $j \geq 1$ and odd $0 < r < 2^j$, this gives

$$4\varphi(2^j n + r) = 2u_{2^j n + r - 1} - u_{2^j n + r}.$$

Thus the $2^e - 1$ independent odd-residue totient sections lie in the span of the carry sections through level e . This explains how the rank lower bound survives even though the sum is rational. The [construction and its limitations](#) explain why it fails the exact prime-dilation laws of φ . The same section retains the cofinal LCM-diagonal criterion, its endpoint sign conditions and the limits of finite integer combinations of shifts.

The [signed interpolation argument](#) tests whether preserving specified analytic data fixes the value. For each integer $\beta \geq 2$, sparse even-index corrections realise every value in an interval about $\sum \varphi(n)\beta^{-n}$, with a prescribed prefix and all odd coefficients fixed. Errors are bounded by any prescribed positive $f(n) \rightarrow \infty$; their support has upper Banach density zero. On each fixed arithmetic progression the Dirichlet-series difference is entire. At roots of unity the generating-function difference extends smoothly radially, with any prescribed finite collection of radial derivatives zero. This preserves Dirichlet singularities, not regular parts or zeros; a finite jet is not the full radial germ. The proof is not formalised in Lean and estimates no cancellation for the unchanged totient sequence.

Rational approximation imposes a different requirement. The identity

$$\Theta_2 := \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2} = S - \frac{1}{2}$$

was posted by Amiram Eldar, together with its coprimality interpretation, on 15 March 2026 [8], and by Steve Fan on 16 May 2026 [9]. Consider the rational quotients

$$Q(e, Y) = \frac{(\sum_{d \leq Y} \mu(d)/(2^d - 1)^{e+2})^2}{\sum_{d \leq Y} \mu(d)/(2^d - 1)^{2e+2}}.$$

Theorem B.3 (A lower bound for the rank-one quotients). *For $e \geq 1$ and $Y \geq 4$, the denominator of $Q(e, Y)$ is positive, and the unique minimiser is $(e, Y) = (1, 5)$. Every admissible quotient and every nonempty finite positive weighted average of such quotients satisfies*

$$Q - \Theta_2 > \frac{21}{320}.$$

The uniform bound $Q - \Theta_2 > 1/15$ is false already at $(1, 5)$.

The [proof](#) first makes the denominator uniformly positive. A geometric bound handles all $e \geq 2$; at $e = 1$, seven-term prefixes leave only $Y = 4, 5, 6$ to compare directly. The averaging assertion concerns averages of the quotients themselves. It leaves open signed combinations, quotients formed after combining the sums, and higher-rank kernels. For a denominator-clearing irrationality argument, a sufficient target is $0 < q_j|\Theta_2 - p_j/q_j| \rightarrow 0$, with p_j/q_j in lowest terms. Indeed, if $\Theta_2 = a/b$, any nonzero difference satisfies $q_j|a/b - p_j/q_j| = |aq_j - bp_j|/b \geq 1/b$. Convergence alone gives no such contradiction.

C An exponential-sum question

For every $h \geq 1$ and every X_0 , suppose that some integers $X \geq \max(X_0, 1)$ and $L \geq h$ satisfy

$$16(2X + h + L + 2) \leq 2^L, \quad \operatorname{Re} \sum_{N=X}^{2X-1} \exp(2\pi i D_{h,N,L}/2^L) \leq \frac{9}{10}X. \quad (6)$$

Only a saving in the real part is needed: one must show that the phases cannot all cluster near 1. If every residue failed (4), each $D_{h,N,L}/2^L$ would be within $1/16$ of an integer. Each exponential would then have real part at least $\cos(\pi/8) > 9/10$. Some N in the block must therefore satisfy (4). For a hypothetical rational value, we choose its period and then a sufficiently late block, contradicting (2).

To separate the counting estimates from the cancellation estimate, we single out one totient in each window. Fix $h, s \geq 1, L \geq s + h, X \geq 1$ and $0 < \eta < 1$, and put $t = L - s + 1$ and $E_N = \exp(2\pi i D_{h,N,L}/2^L)$. For $X \leq N < 2X$, let p_N be the largest prime factor of $N + t$ and $m_N = (N + t)/p_N$. Define

$$\begin{aligned} \mathcal{A} &= \{N \in [X, 2X) : m_N \leq \lfloor \sqrt{X}/2 \rfloor, p_N > 2\lfloor \sqrt{X} \rfloor\}, \\ \mathcal{G} &= \{N \in \mathcal{A} : \varphi(m_N) \geq \eta m_N\}. \end{aligned}$$

All indices are integers. On $\mathcal{A}$, $p_N > m_N$, hence $\varphi(N + t) = \varphi(m_N)(p_N - 1)$. Since $h < t \leq L$, this term occurs in both tails, with net coefficient $2^{-(t-h)} - 2^{-t} = (2^h - 1)/2^t$. Write

$$z_N = \exp\left(\frac{2\pi i (2^h - 1)\varphi(m_N)(p_N - 1)}{2^t}\right), \quad w_N = E_N/z_N,$$

and let $\bar{z}_m$ be the average over the indices in $\mathcal{A}$ with cofactor m , taking the average to be zero for an empty set. We seek the four bounds

$$\begin{aligned} \operatorname{Re} \sum_{N \in \mathcal{G}} w_N (z_N - \bar{z}_{m_N}) &\leq \frac{14}{25}X, \\ \left| \sum_{N \in \mathcal{G}} w_N \bar{z}_{m_N} \right| &\leq \frac{1}{100}X, \\ \left| \sum_{N \in \mathcal{A} \setminus \mathcal{G}} E_N \right| &\leq \frac{1}{100}X, \\ \left| \sum_{\substack{X \leq N < 2X \\ N \notin \mathcal{A}}} E_N \right| &\leq \frac{8}{25}X. \end{aligned} \quad (7)$$

These are an exact partition: outside $\mathcal{G}$ use the last two sums; on $\mathcal{G}$ write $w_N z_N = w_N (z_N - \bar{z}_{m_N}) + w_N \bar{z}_{m_N}$. The bounds add to $(14/25 + 1/100 + 1/100 + 8/25)X = 9X/10$. There are two distinct issues. Small group means control the second sum by the triangle inequality because $|w_N| = 1$. They do not control the first: subtracting $\bar{z}_m$ cancels an *unweighted* group sum, not one with varying w_N . No independence is assumed. For example, $z = (1, -1)$ has mean zero, but the weights $w = (1, -1)$ make $\sum w_N (z_N - \bar{z}) = 2$.

This illustrates the distinction, not the behaviour of the totient phases. The resulting real-part bound is not a $21X/25$ norm bound.

Problem C.1 (Four estimates for the first-harmonic sum). For each $h \geq 1$, do there exist $s \geq 1$ and $0 < \eta < 1$ such that, for every X_0 , there are X, L with

$$\max(X_0, 1) \leq X, \quad h \leq L - s, \quad 16(2X + h + L + 2) \leq 2^L,$$

for which those four bounds hold?

Fix s, η after h but before X_0 ; all four estimates must hold on the same block. Take the least $L \geq s + h$ satisfying (6)'s size inequality. The [smooth-number and small-cofactor estimates](#) give the last two bounds for all sufficiently large X when $0 < \eta < 1/(300 \log 4)$. The elementary excluded-cofactor estimate is $(3\eta \log 4 + o(1))X$; it needs no prime number theorem. With the prime number theorem as input, the [group-mean estimate](#) gives the second bound for $s = 26$, $\eta = 1/1000$ at this depth and all large X ; the mean-subtracted bound remains open. At these fixed parameters it would suffice to prove that bound for arbitrarily large X : the other three already hold beyond fixed thresholds and hence on those same sufficiently late blocks.

There is also a weaker sufficient condition, without a group-mean input. Since $|\mathcal{G}| > 67X/100$ for large X , the open bound $\text{Re} \sum_{N \in \mathcal{G}} E_N \leq 603X/1000$ would put the average on $\mathcal{G}$ below $9/10$: $603/1000 = (9/10)(67/100)$. Some good index would then pass the test. This uses an average on $\mathcal{G}$, not a bound on the full block, and is required for each h on arbitrarily large blocks. The [ordinary comparison argument](#) gives larger first harmonics at all sufficiently large admissible scales, distinguishing its rational sequences from the cancellation required of φ . Its proof is included with the interpolation construction.

Balasubramanian, Giri and Srivastav give correlation estimates applicable to $\varphi(n)/n$, uniformly for $|h| \leq x/2$ [11, Thm. 2.2]. Their correlation is a product of two normalised totient values. Here the modulus grows with L , and w_N is an exponential of the other, overlapping totient contributions. Uniformity in the shift of that two-point product does not supply the required weighted exponential estimate.

Verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The [declaration guide](#) links the formal sources for this paper. Complete proofs of three cited results appear in Appendix A of the companion paper. The displayed arguments, including signed interpolation, require ordinary mathematical review; the recorded Lean and Comparator checks cover their specified declarations and source versions. No new formal run or independent review was performed for this revision.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.1	Lean · Comparator	Corollary 3.1	Lean · Comparator
Theorem 4.1	Lean · Comparator	Lemma 4.2	Lean · Comparator
Corollary A.1	Lean · Comparator	Corollary A.2	Lean · Comparator
Theorem B.1	Lean · Comparator	Theorem B.3	Lean · Comparator